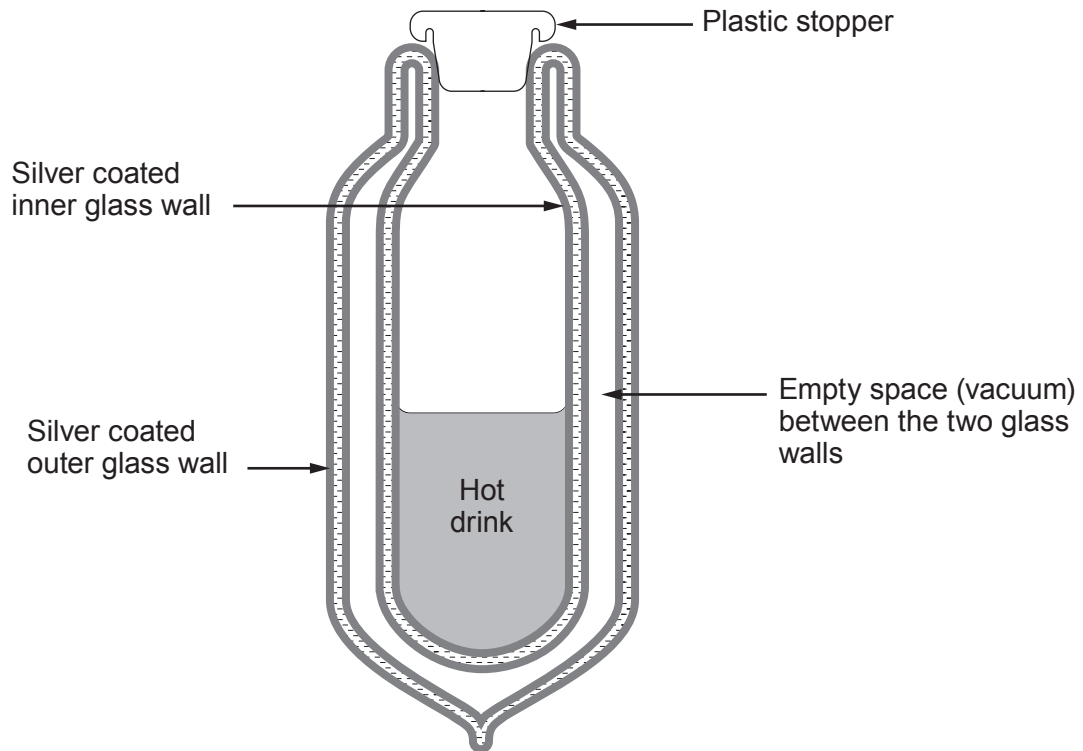


Examiner
only

6. On a cold day a vacuum flask can be used to keep drinks hot for many hours. The diagram shows a vacuum flask with its main features labelled.



- (a) Explain in terms of conduction, convection and radiation how the labelled features of the flask help to keep the drink hot for such a long time. [6 QER]

- (b) A drink at 90 °C is poured into the flask. After 2 hours the temperature of the drink is 80 °C. After 10 hours the temperature of the drink is 60 °C.

The company who make the vacuum flask claim that a hot drink contained in the flask will cool down by 5 °C every hour. Explain if the claim made by the company is always correct. [2]

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